

Overview of ASME V&V 20-2009 Standard for Verification and Validation in Computational Fluid Mechanics and Heat Transfer

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History of V&V 20



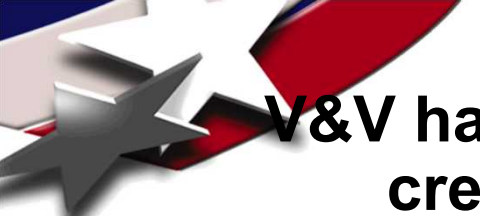
- In 2004, Professor Hugh Coleman, an expert on experimental uncertainty analysis, sought out a collection of V&V people
- Purpose was to provide an approach to validating (to be defined) computational simulations
- Academia, industry and national labs were represented on the committee
- Techniques for developing uncertainty analysis for computational simulations should be consistent with those used in experimental community (ASME PTC 19.1 and ISO GUM)



V&V 20 Committee



- H. W. Coleman (chair), U. Alabama/Huntsville
- C. J. Freitas (vice chair), SWRI
- R. L. Crane, ASME
- B. F. Blackwell, Consultant
- K. J. Dowding, Sandia
- U. Ghia, U. Cincinnati
- R. G. Hills, Sandia
- R. W. Logan, Consultant
- P. J. Roache, Consultant
- W. G. Steele, Jr., MSU



V&V has the goal of providing evidence for the credibility of computational simulation



- Computational simulation is (relatively) inexpensive and has advanced to the point that complex systems can be analyzed
- While computational simulation can be applied to complex systems, the accuracy of the results are often unknown
 - Computer codes to solve the simulation models may have bugs or algorithm deficiencies (code verification)
 - Computational simulations approximate the solution of the mathematical equations (solution verification)
 - » Discretization dependencies
 - The mathematical equations solved in a computational simulation (PDEs usually) are an approximation at some level (validation)
 - » May be based on fundamental principles, but typically contain constitutive models or “sub-grid” models that are empirically based
 - » There are simplifications and assumptions in developing the computation simulation (validation)



Definitions of Verification



- Verification
 - The process of determining that a computational model accurately represents the underlying mathematical model and its solution. (V&V-10, high level document for solid mechanics)
 - Code verification (V&V-20): “Establishes that the code accurately solves the mathematical model incorporated in the code, i.e. that the code is free of mistakes for the simulations of interest.”
 - Solution verification (V&V-20): Estimates the numerical accuracy of a particular calculation.
- Code Verification precedes validation
 - Establishes that the code solutions are an accurate representation of the mathematical models
 - Goal is finding and removing mistakes in the source code and numerical algorithms
- Solution (Calculation) Verification performed as part of validation
 - Assuring the numerical accuracy of code output for the problem of interest



Definition of Validation



- Definition Validation (ASME V&V-10/V&V-20):
“The process of determining the degree to which a model is an accurate representation of the real world from the perspective of the intended uses of the model.”
- Key points:
 - “Process” (that may be iterative)
 - “degree .. accurate representation of real world” (quantitative)
 - Intended uses (adequacy)

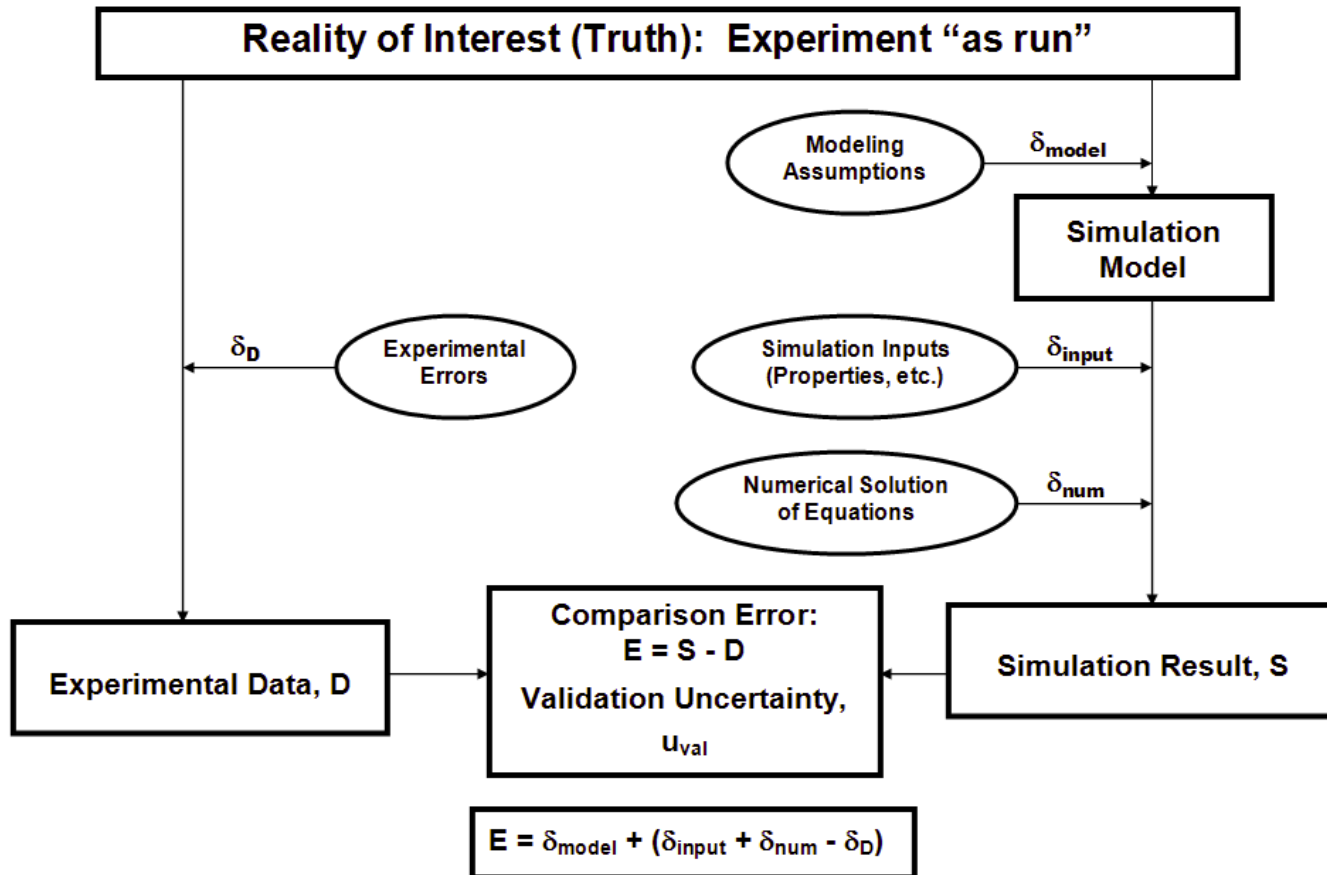


Code Verification



- Code verification should be performed prior to validation
- Establishes that the code solutions are an accurate representation of the mathematical equations in the simulation
 - Goal is finding and removing mistakes in the source code and numerical algorithms
- Performed by comparing analytical solutions of mathematical equations to simulation code solutions
 - Method of Manufactured solutions (MMS) (see Roach, 1997 or Knupp and Salari, 2003)
 - Classical analytical solutions (Beck)
- Simple (conceptually) for single physics (equation) models
 - NS for Laminar flow, Nonlinear heat conduction, etc.
- Challenges exist for verifying the fully-coupled mathematical equations
 - CFD for turbulent reacting flow with participating media
 - Nonlinear heat conduction with enclosure radiation and chemical kinetics

Overview of the V&V-20 Validation Procedure





Detailed Description of V&V-20 Methodology (4)



- Provided the errors are independent

$$u_{val} = \sqrt{u_{num}^2 + u_{input}^2 + u_D^2}$$

- u_{num} – uncertainty due to numerical solution of the equations – estimated with code verification and solution verification processes
- u_{input} – uncertainty due to simulation input uncertainty – estimated with techniques discussed in this course
- u_D – uncertainty due to experimental measurement uncertainty – estimated with similar techniques
- Methodology is provided in V&V-20 for the case when errors are not (approximately) independent
 - Effect of assuming independence in the errors is problem specific



Numerical solution uncertainty (u_{num})



- Code verification needs to be conducted for the simulation capabilities used in a validation activity
 - The evidence for trusting that the simulation code is accurately solving the mathematical equations
 - Code verification does not directly contribute to u_{num}
- Numerical uncertainty u_{num} estimated with Grid convergence index (GCI)
- Requires running the simulation model while refining the grid (mesh)
- Procedures to estimate an uncertainty that bounds the numerical effects is derived directly from the GCI



Simulation input uncertainty (u_{input})



- Both Mean Value (MV, propagation of variance) and Latin Hypercube Sampling (LHS) approaches are described for propagating simulation input uncertainty through the simulation software
- Provides an estimate of u_{input} that characterizes a range of the simulation that can be attributed to input uncertainty



Experimental uncertainty (u_D)



- Based on the approaches described in ASME PTC 19.1 (Test Uncertainty) and ISO GUM
- Makes a distinction between random and systematic contributors to the experimental uncertainty
- Both Mean Value (MV) and Latin Hypercube Sampling (LHS) are discussed in V&V 20



Validation Metrics



- Validation metrics involve selecting the what(s) and how(s)
- What can be compared in the validation experiment?
 - Validation at the lower levels measures responses that are not easily relatable to validation responses at higher levels
- How will the comparisons be quantified
 - Statistics of the measurement-simulation differences
 - Comparison of the measurement-simulation differences to uncertainty
 - Estimate the magnitude of model form error



V&V-20 Definitions



- Definitions of Error and Uncertainty*
 - (δ) error (of measurement): “result of a measurement minus a true value of the measurand”, and
 - (u) *uncertainty (of measurement)*: “*parameter, associated with the result of a measurement, that characterizes the dispersion of the values that could reasonably be attributed to the measurand*”.
- An error δ is then a quantity that has a particular sign and magnitude
 - Assume that any known errors have been removed
 - Remaining error is thus of unknown sign and magnitude
- Uncertainty u is *estimated with the idea that $\pm u$ characterizes the range containing the error δ* .
 - Consider u an estimate of the standard deviation

* International Vocabulary of Basic and General Terms in Metrology, 2nd Edition, International Organization for Standardization, Geneva, Switzerland, 1993.



Detailed Description of V&V-20 Methodology (1)



- Simulation (S) and experimental data (D) are compared

$$E = S - D$$

- Simulation error

$$\delta_S = S - \mathcal{T}$$

- Experimental error

$$\delta_D = D - \mathcal{T}$$

- Comparison Error

$$E = S - D = (\mathcal{T} + \delta_S) - (\mathcal{T} + \delta_D) = \delta_S - \delta_D$$



Detailed Description of V&V-20 Methodology (2)



- Simulation error has 3 components

$$\delta_S = \delta_{model} + \delta_{num} + \delta_{input}$$

- δ_{model} is due to modeling assumptions and approximations
 - δ_{num} is due to numerically solving the model equations
 - δ_{input} is due to error in the simulation input parameters
- Comparison error can be written in terms of all contributions

$$E = S - D = \delta_{model} + \delta_{num} + \delta_{input} - \delta_D$$

- Rearranging gives an expression for the modeling error

$$\delta_{model} = E - (\delta_{num} + \delta_{input} - \delta_D)$$



Detailed Description of V&V-20 Methodology (3)



- Errors are of unknown sign and magnitude
- Consider estimating an uncertainty to characterize the range containing the error
- *Validation standard uncertainty, u_{val}*
 - standard deviation of the parent population of the combination of errors ($\delta_{num} + \delta_{input} - \delta_D$)
- Given the validation uncertainty, the interval

$$E \pm u_{val}$$

- Characterizes a range within which δ_{model} falls

$$\delta_{model} \in [E - u_{val}, E + u_{val}]$$

- Estimation of u_{val} is at the core of the methodology presented in V&V-20
 - E and u_{val} are the validation metrics



Summary



- V&V-20 provides a methodology to estimate the contribution for sources of uncertainty related to comparing a measurement and simulation model
 - Numerical solution uncertainty
 - Simulation input parameter uncertainty
 - Experimental measurement uncertainty
- An approach for aggregating the uncertainties and measuring the measurement-simulation difference relative to the total uncertainty is provided
- Guidance on interpreting of the comparison is given and a procedure to characterize model form error – due to approximations and assumptions in the simulation – is discussed



Backups



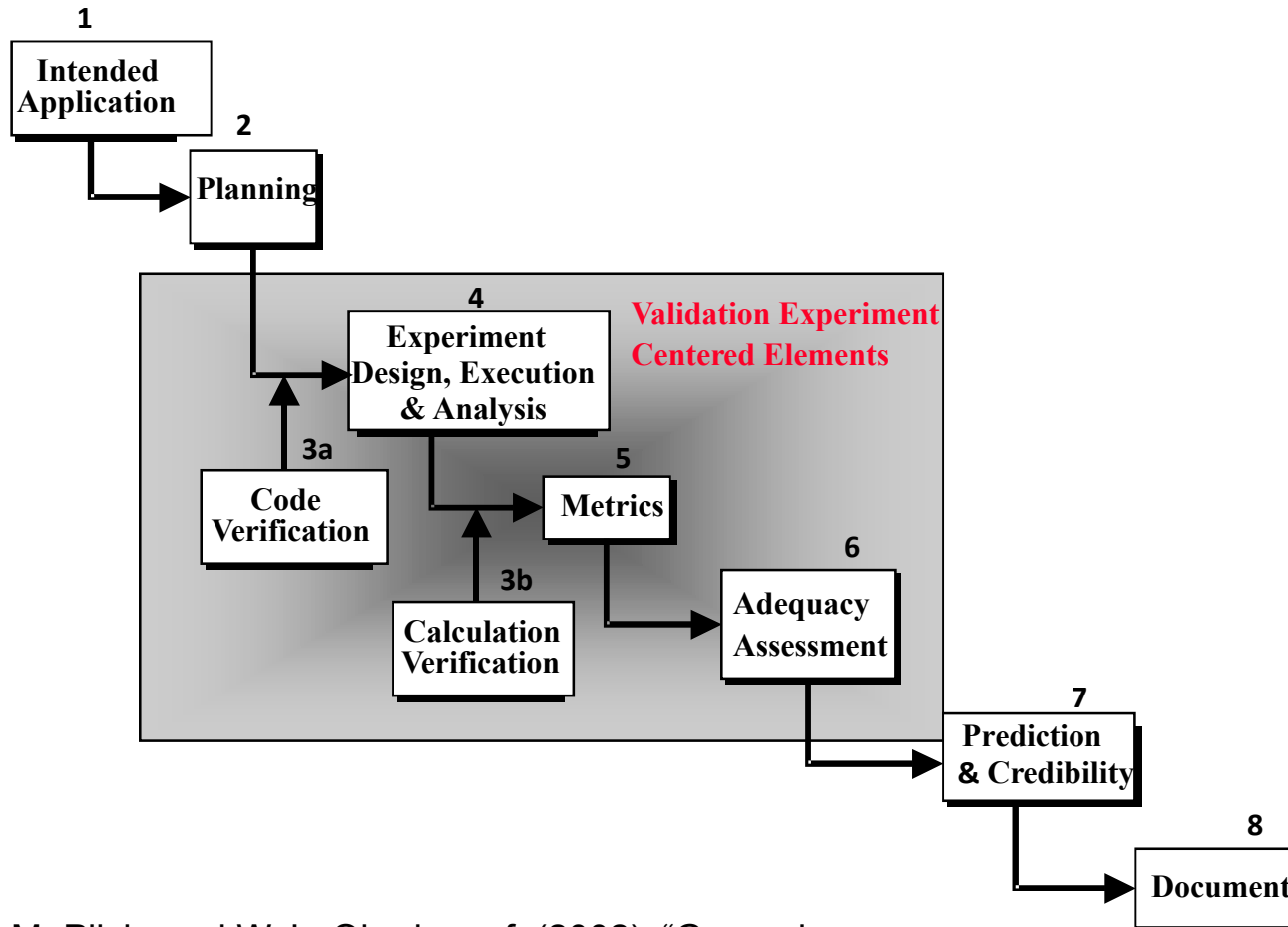


Outline



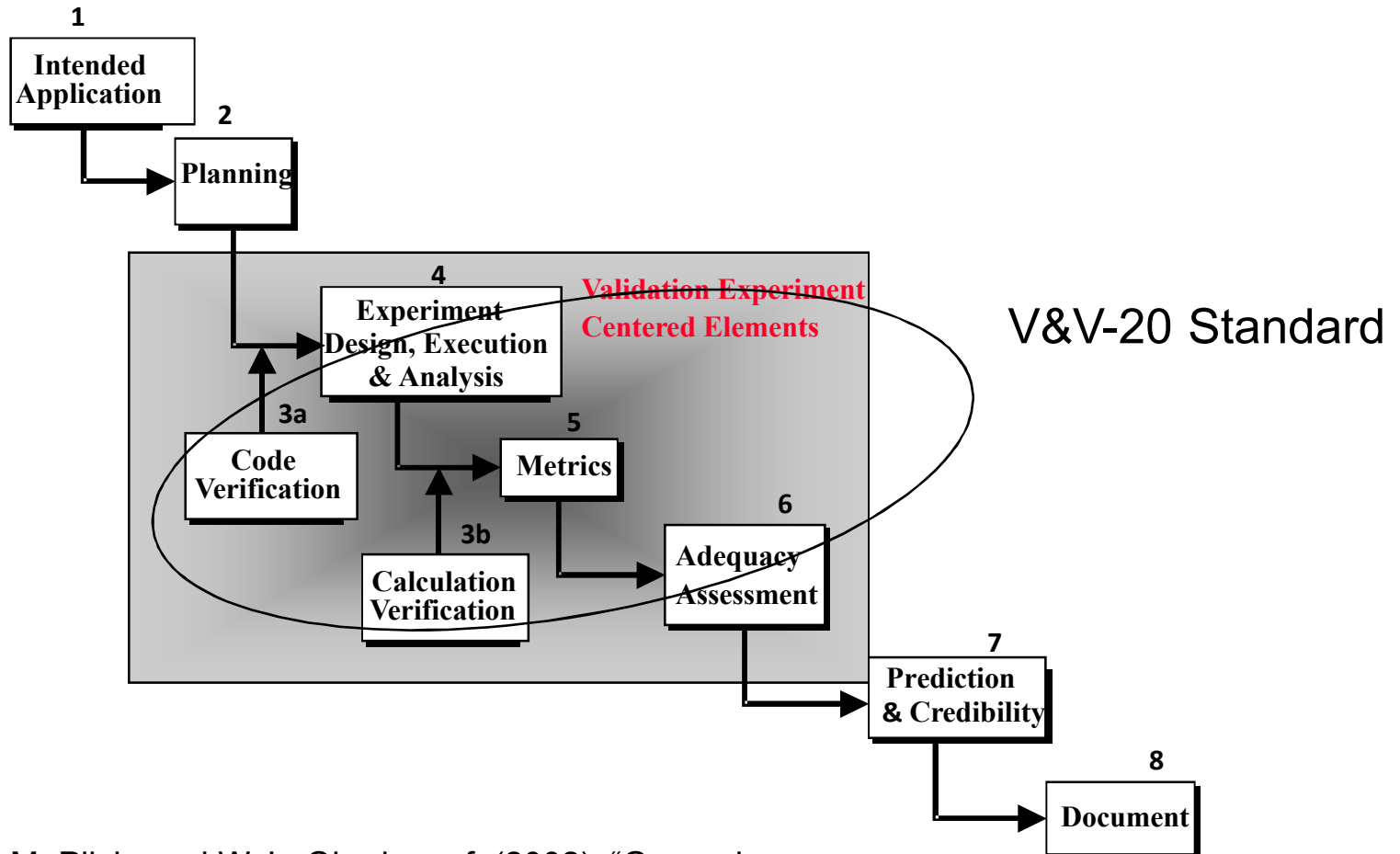
- **Verification and Validation Concepts/Process**
 - Verification and Validation Definitions
 - Overview of the V&V Process high-level steps
- **Summary of V&V Standard from ASME V&V20**
 - Scope and conceptual methodology
 - Description of steps in the methodology
 - Interpretation of the validation comparison
 - Summary of methodologies for addressing uncertainties

Overview of the Validation Process



Trucano, T. G., M. Pilch, and W. L. Oberkampf, (2002), "General Concepts for Experimental Validation of ASCI Code Applications," report SAND2002-0341, Sandia National Laboratories, Albuquerque, NM, March 2002.

Overview of the Validation Process



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2. Planning

- Given the sheer number of capabilities in a code, the V&V activities need to be prioritized
 - Thermal analysis code has 120 capabilities and features
 - Subset is used in certain classes of problems
- Phenomenon Identification and Ranking Tables (PIRT) are one option to prioritize the V&V activities for an application
 - From the Reactor Safety Community
 - Table identifies and ranks the Phenomena based on expert opinion

	Importance	Model	Code Ver.	Validation
Phenomena 1	1 (high)	3	2	0
Phenomena 2	3 (low)	1	1	2
Phenomena N	2	2	0	0



4. Design, Execution, and Analysis



For a complex application validation experiments conducted at various levels of physics and geometric fidelity

- Constitutive model validation (material characterization) (10 – 100 of exps)
 - Representative of intended application environment and materials
- Ensemble Validation (single physics/simple coupling and stylized hardware) (10s of experiments)
 - Study accuracy over a range of the conditions
 - May do this for several of important single physics
- Subsystem Validation (Dominant physical coupling and representative but simplified hardware) (> 10s of exps)
- Full system Validation (full coupling with actual or slightly simplified hardware) (1 exp)
- Designing the suite of experiments is a challenging activity
- “Top-Down” or “Bottom-up” are typical hierarchal validation approaches



6. Adequacy Assessment



- Is the measured accuracy (value of the metric) adequate for the intended application??
- Judging model adequacy at lower levels in the validation hierarchy can be difficult
 - Measured quantities at lower levels may be related, but not directly applicable to higher levels and the intended application
 - Lower level validation may be a more highly stressing application of the model than at higher levels
- One approach is based on measuring the comparison relative to uncertainty
 - Judge if the simulation model has adequate uncertainty relative to the exp data -> if so, the simulation model including all uncertainties is adequate for application at higher levels and for an intended application



Intended Application

- Goal of validation is to validate the simulation model for uses other than where validation is performed (an intended application)
- The intended application needs to be carefully considered. Questions to consider:
 - What is the application of interest to apply the simulation model?
 - What simulation outputs are needed and for what simulation environments?
 - » Lift/drag, maximum temperature, temperature difference, incident heat flux, etc.
 - » Subsonic/supersonic, shock loading, Fire-level fluxes
 - What is the required accuracy?
 - » 20%, factor of 2, conservative
- Clear line-of-sight back to the intended application is needed for a validation activity to be successful
 - Success doesn't necessarily require the model to be acceptable



7. Prediction and Credibility



- How are the results of the validation experiments aggregated in the prediction
 - Model of “Best Estimate + Uncertainty” for making a prediction in an application
- Credibility is aimed at due diligence in addressing the various steps in the overall process (what is the level or rigor achieved)
 - Coverage of the code verification test suite
 - Validation hierarchy is complete
 - Rigorous validation metrics used that address uncertainties
 - Etc.



How do we move beyond the (validation) process level??



- ASME V&V-20 committee developed a standard for “Verification and Validation in Computational Fluid Dynamics and Heat Transfer” (due late summer/early fall)
- Standard is based on definitions of error and uncertainty that have been internationally codified by the experimental community over several decades
- Guide to the Expression of Uncertainty in Measurement (GUM) was published by the International Organization for Standardization (ISO) in its name and those of six other international organizations.
 - Bureau International des Poids et Mesures (BIPM), International Electrotechnical Commission (IEC), International Federation of Clinical Chemistry (IFCC), International Union of Pure and Applied Chemistry (IUPAC), International Union of Pure and Applied Physics (IUPAP), and International Organization of Legal Metrology (OIML)



Validation methodology developed using error and uncertainty concepts



- These (error and uncertainty) concepts were first extended in [Coleman and Stern, JFE 1997] for Validation
- V&V-20 Validation Methodology
 - Scope was to quantify the degree of accuracy inferred from the comparison of solution (simulation) and data for a specified variable at a specified validation point
 - Approaches for estimating the uncertainty in the experimental response (based on ASME PTC 19.1)
 - Approaches for code and solution verification (Method of Manufactured Solutions, MMS, and Grid Convergence Index, GCI)
 - Approaches for estimating uncertainty due to uncertainty model inputs (MV and LHS)
 - Metric for comparing simulation and data while including all previously mentioned uncertainties



Interpretation of Validation Results (1)

- V&V-20 methodology provides the following information without making any assumptions on the parent population of combination of errors

$$(\delta_{num} + \delta_{input} - \delta_D)$$

$$\delta_{model} \in [E - u_{val}, E + u_{val}] = E \pm u_{val}$$

- Two cases can be considered:

1. $|E| \gg u_{val}$

then $\delta_{model} \approx E$

2. $|E| \leq u_{val}$

then δ_{model} is of the same order, or less than, the combination of errors $(\delta_{num} + \delta_{input} - \delta_D)$



Interpretation of Validation Results (2)



- To identify an interval within which δ_{model} falls for a given probability level requires a distribution for the parent population of the combination of errors ($\delta_{num} + \delta_{input} - \delta_D$)
- Given a probability distribution a coverage factor, k , can be selected

$$U_{\%} = k u$$

- U is the expanded uncertainty, U_{95} is an interval within which the error would fall about 95 out of 100 times

$$\delta_{model} \in E \pm k u_{val}$$

- Uniform, $k=1.73$ for 100% of the population
- Triangular, $k=2.45$ for 100% of the population
- Normal, $k = 3.00$ for 99.7%, $k = 3.5$ for 99.95%



ASME



- **ASME overview by Ryan Crane**



Origins of The Society

- Technical professional society; founded in 1880 as the American Society of Mechanical Engineers; not-for-profit
- Engineers convened to foster engineering innovation, education and research
- First standard issued in 1884 – boiler efficiency test
- Headquarters in New York City
- Offices in Atlanta, Houston, New Jersey, Beijing (China), Brussels (Belgium), Mumbai (India)



Origins of The Society

ASME's Mission:

- To serve our diverse global communities by advancing, disseminating and applying engineering knowledge for improving the quality of life; and communicating the excitement of engineering



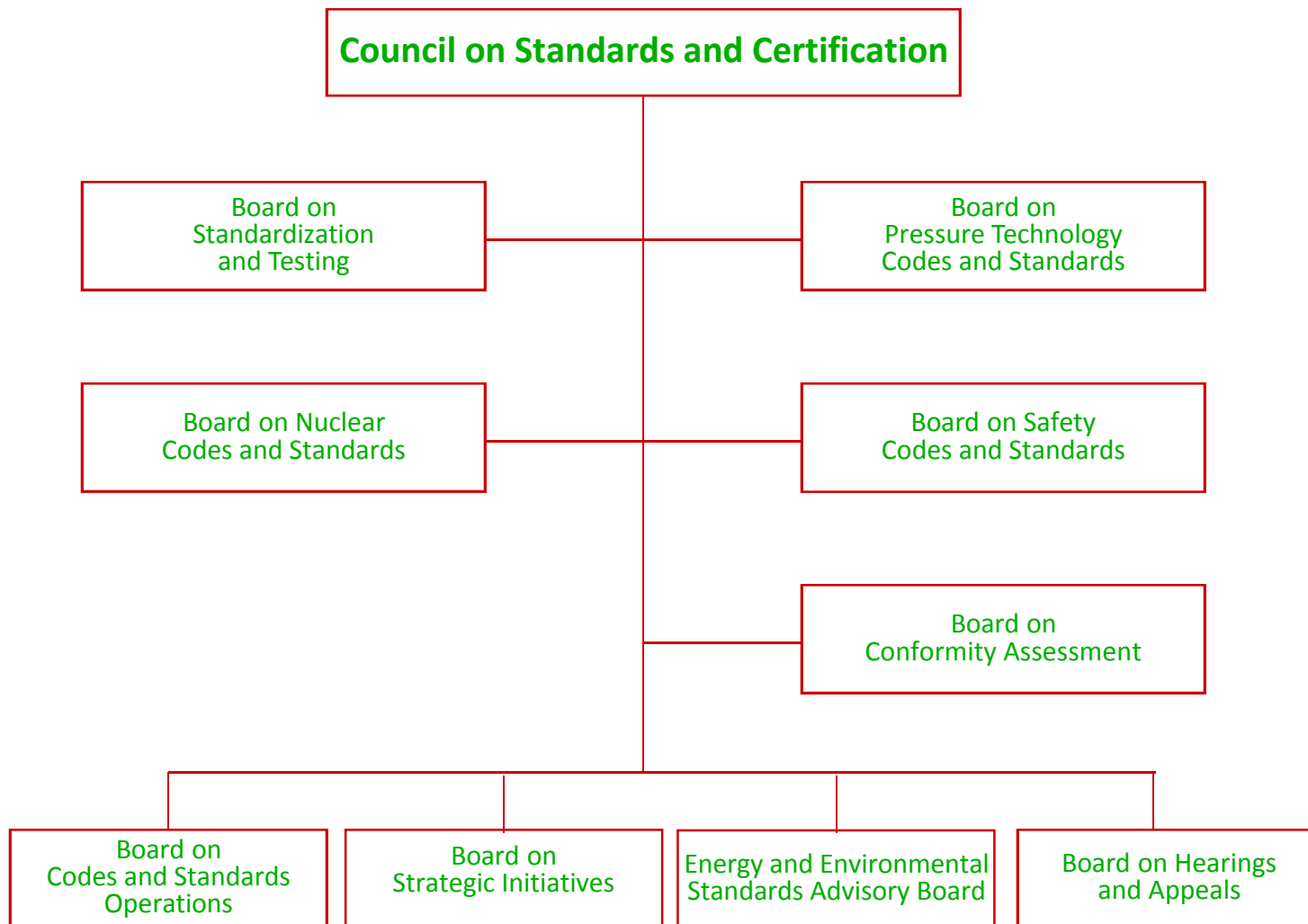
ASME Standards and Certification



- 70 Consensus Committees
- 700 Total Committees
- 5 Supervisory Boards
- 5,100 Volunteers (800 International and growing)
- 550 Standards
- ASME Standards Accepted for Use in Over 100 Nations
- Administer Over 40 U.S. TAG's to ISO
- 12 Conformity Assessment Programs
- More than 200 training programs worldwide



Standards & Certification Organization





ASME Codes and Standards Consensus Process



- Decisions are reached through *consensus* among those affected.
- Participation is *open* to all affected interests: no membership fees or requirement to be ASME society member
- *Balance* is maintained among competing interests
 - Designers, Manufacturers, Users, Regulators, General Interests, etc.
- The process is *transparent* - information on the process and progress is directly available.



ASME Codes and Standards Consensus Process



- ***Due process*** assures that all views will be considered and that appeals are possible.
- The process is ***flexible***, allowing the use of different methodologies to meet the needs of different technology and product sectors.
- The process is ***timely***; purely administrative matters do not slow down the work.
- Standards activities are ***coherent***, avoiding overlap or conflict.
- Procedures accredited by the American National Standards Institute (ANSI)
- Consistent with Principles of WTO Technical Barriers to Trade Agreement



What does "Consensus" mean?

- **ASME standards are consensus standards**
- **Consensus means**
 - Substantial agreement by affected interest categories
 - Consideration of views, attempted resolution is part of the process
 - Unanimity not required
- **The Standards Committee is responsible for establishing consensus on proposed standards actions.**



The Consensus Development Process

- **Development of standards action**
- **Recorded vote**
- **Public review**
- **Supervisory Board approval**
- **Appeals**
- **ANSI approval**



ASME Codes and Standards

Proprietary Issues

- **ASME does not ‘approve’, ‘rate’ or ‘endorse’ any item, construction, proprietary device, or activity**
- **Proprietary devices may only be incorporated in a generic manner**



Participation



Volunteer Participation

- Individual “voluntary” participation
- ASME Codes & Standards relies on industry supporting participation by knowledgeable experts
 - Involve the best and the brightest people from all around the world to develop, maintain, and promote the use of these ASME products and services
- ASME membership is not a prerequisite for committee membership
- Technical and administrative support to the committees
- International Working Groups in China, Germany, India, Italy, Korea, and Pan-European Union



ASME Codes and Standards Development

- **No funding by government**
- **No fees to participate**
- **Supported by revenue from S&C sales and related programs**
- **Volunteer members not compensated by ASME**



U.S. Public Policy

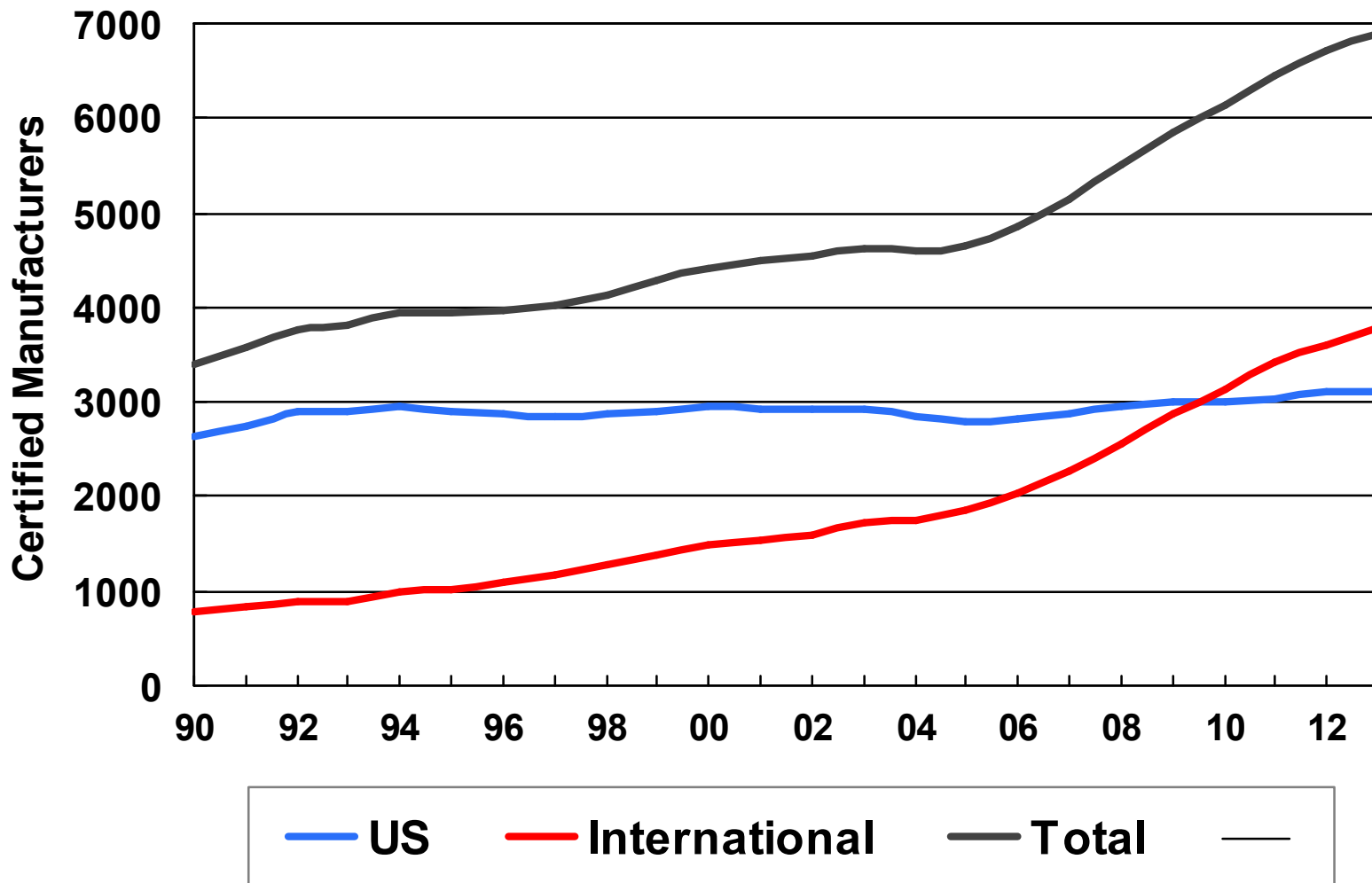


- OMB Circular A119
 - Instructs Federal Agencies to use voluntary standards and participate in their development, unless impractical
- National Technology Transfer and Advancement Act of 1995
 - Mandates government agency use of private sector voluntary consensus standards
 - Obliges federal government participation in voluntary standards development
- Fulfills Regulators' Expectations
 - Meets public safety, health and environmental objectives
 - Provides confidence in the technical integrity of the product or service
 - Supports mutual recognition essential to international trade



ASME Global Growth

Certified Boiler & Pressure Vessel Manufacturers





Joint Efforts



- API 579-1/ASME FFS-1- Fitness-for Service
- Joint ACI-ASME Committee on Concrete Components for Nuclear Service
- ISO/ASME 14414 - Pump system energy assessment
- ASTM material specifications for ASME Section II

- SDO Convergence Board
- Input to MDEP CSWG and Code Comparison Project
- Nuclear Energy Standards Coordination Collaborative (NESCO)
- WNA Cooperation in Reactor Design Evaluation and Licensing (CORDEL)
- NAFEMS Analysis Management Working Group



Verification and Validation Standards Committee



- Four Subcommittees on Verification and Validation in Computational/Computer
 - V&V 10 - Solid Mechanics
 - V&V 20 - Fluid Dynamics and Heat Transfer
 - V&V 30 - Simulation of Nuclear System Thermal Fluids Behavior
 - V&V 40 - Modeling of Medical Devices
 - V&V 50 - Advanced Manufacturing
 - Others?